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Pre-Service Teachers' Interpretations and Decisions About a 3D Geometry Activity Sequence

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Abstract

The most widespread lesson preparation resource used by mathematics teachers is the textbook. Initial teacher training programmes should therefore develop the skill of curricular noticing, i.e., the ability to critically analyse and make decisions concerning an activity sequence from a textbook. This mix-method study focused on the interpretations and decisions adopted by 85 Spanish pre-service primary school teachers (PTs) in relation to a three-dimensional (3D) geometry activity sequence from a textbook. The PTs were assigned two tasks: the first was identifying the limitations of the activity sequence for supporting students' geometrical understanding regarding three aspects—attributes, geometrical processes, and modes of representation—and the second was completing the sequence. Most PTs interpreted a number of activity sequence limitations. In terms of their decision-making, the PTs prioritised certain characteristics over others, such as introducing further attributes rather than changing representation modes, or adding geometrical processes to their activity sequence designs. Moreover, the analysis allowed determining how PTs completed the activity sequence to address limitations, thereby revealing relationships between their interpretations and decisions. The study findings help teacher educators to design courses aimed at supporting the PTs' ability to make more informed and effective teaching choices that enhance student learning.

Keywords: curricular noticing; pre-service primary school teachers; textbooks; 3D geometry

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Academic Editor: Susanne Garvis

Received: 11 November 2025
Revised: 12 December 2025
Accepted: 29 December 2025
Published: 31 December 2025

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1. Introduction

In the last two decades, professional noticing competence has become a central research subject within mathematics education (e.g., recent reviews or special issues in Dindyal et al., 2021; Weyers et al., 2024; Zhang et al., 2025). The reviews indicate that most studies have centred on the noticing of students' mathematical thinking, a construct conceptualised by Jacobs et al. (2010) as three core skills: attending to mathematical details in students' strategies; interpreting students' mathematical thinking; and deciding how to support that thinking. Such research has shed light on how pre-service or in-service teachers attend to, interpret or decide about students' mathematical thinking across different mathematical domains (e.g., Callejo & Zapatera, 2017; Gupta et al., 2018; Moreno et al., 2021). It also provides tools and contexts to develop this competence in initial teacher education programmes or professional development programmes (Amador et al., 2017; Fernández et al., 2018; Ivars et al., 2020; Jacobs et al., 2024). According to these latter

studies, decision-making is the hardest and most demanding skill to acquire. Teachers and pre-service teachers can indeed attend to and interpret specific details, but they have difficulties in integrating these interpretations into the decisions they take regarding the optimal teaching steps to adopt next (Barnhart & van Es, 2015; Ivars et al., 2020; Rotem & Ayalon, 2024).

The difficulties that teachers or pre-service teachers encounter when designing activities may be due to the materials and resources available to them and the challenges they present. It is thus relevant to focus on how pre-service teachers interact with curricular materials. Such curricular materials are wide-ranging: from written and digital textbooks, teaching programmes and webpages to manipulative resources. More specifically, textbooks are the most frequently used resource in lesson preparation and therefore play a significant role in many educational systems (Blanco et al., 2021; Fan et al., 2013; Glasnovic Gracin, 2018). However, shortcomings have been identified in the way textbook contents are presented (Girit-Yildiz & Ulusoy, 2024). Teachers must therefore implement these materials thoughtfully: they need to identify both their affordances and their limitations with respect to meaningful mathematical learning (Choy & Dindyal, 2021).

In this context, it is important to explore how teachers develop their curricular noticing competence. Curricular noticing refers to a teacher's ability to make sense of the content complexity as well as the pedagogical opportunities present in curricular materials (Amador et al., 2017; Dietiker et al., 2018).

The present study centred on how a group of pre-service Spanish primary school teachers noticed an activity sequence in a textbook. The activity sequence addressed three-dimensional (3D) geometry which is a pivotal subject in the primary school curriculum of many countries. Specifically, 3D geometry is introduced in Year 1 (students aged 6 to 7 years) in Spain. Previous research has not only identified limitations in textbook geometry contents (e.g., Glasnovic Gracin, 2018; Purnomo et al., 2024) but also the difficulties of pre-service teachers and teachers in understanding and teaching geometry concepts (e.g., Aslan-Tutak & Adams, 2015; Shayeb et al., 2025; Sunzuma & Maharaj, 2019). The section deepens these two latter points based on a literature review.

2. Literature Review and Theoretical Framework

First, we review the literature on textbook geometry contents and on pre-service teacher geometry knowledge. We then present the conceptualisation of curricular noticing competence as well as the study objective and specific research questions.

2.1. Literature Review on Textbooks: Geometry

Geometry textbook research has centred, among others, on activities—in terms of type of context—geometrical processes (i.e., the different types of thinking required to perform the task), and representation modes—. Concerning the type of context, Mthethwa et al. (2024) found that Euclidean geometry activities in South African textbooks offered few opportunities for engaging students in real-life scenarios, presenting instead numerous intra-mathematical activities but no practical applications. This pattern was also identified in geometry activities drawn from Croatian, Taiwanese, Singaporean, Finnish and American textbooks, where the activities were disconnected from real-life scenarios (Glasnovic Gracin, 2018; Yang et al., 2017). A study of Indonesian and Singaporean textbooks on the angle concept further confirmed this deficiency (Purnomo et al., 2024).

With respect to geometrical processes, Purnomo et al. (2022) reported that Indonesian textbooks mostly promote the application of calculation routines by requiring students to determine perimeters, areas, and volumes by merely directly applying formulas: few opportunities are thus provided to analyse geometrical elements and constructs. The latter

conclusions support the findings of Glasnovic Gracin (2018) on Croatian textbooks. By encouraging such mechanical application, students fail to develop their ability to make conjectures, reason, or generalise, which are all essential skills for solving complex 2D and 3D geometry activities. Added to application activities, Chilean textbooks (Díaz-Levicoy et al., 2019)—particularly those designed for earlier school years—integrate a large number of exercises directed towards identifying properties or shapes, rather than towards promoting deeper conceptual understanding.

On the subject of modes of representation, Yang et al. (2017) found that Taiwanese textbooks tended to combine various representation modes within each activity, while Finnish and American geometry activities generally adopted a single representation. Specifically, Finnish geometry activities rely mainly on graphical representations (images, diagrams, charts, and other visual elements), while American ones include either graphical or written representations (definitions or activity statements). However, Indonesian activities (Purnomo et al., 2024) primarily involved a pure mathematical form of representation by integrating mathematical expressions only (such as algebraic equations). Focusing on graphical representations, Barrantes-López et al. (2014) conducted an in-depth analysis of 3D geometric figures in Spanish textbooks. Specifically, they encountered that most graphical representations of prisms, pyramids, cones and cylinders were depicted vertically standing on their base (a prototypical position). The predominance of prototypical figures, together with the limited variety of representation modes, can create highly rigid images in students' minds that hinder their ability to recognise a figure when it is presented in a non-prototypical position. Furthermore, Barrantes-López et al. (2014) identified a lack of relevant elements in the representations of 3D geometrical figures, such as hidden edges, axes of symmetry, and auxiliary lines. This absence prevents many students from visualising the complete structure of figures, leading to conceptual gaps that may hinder geometrical processes such as analysing and classifying.

The limitations mentioned above highlight the need for teachers to adopt a critical approach whey they use textbooks, particularly in geometry. Analysing textbook affordances and limitations can help teachers attend to and interpret conceptual or procedural gaps or biases in content presentation, allowing them to complete, reformulate, modify, or design effective activity sequences.

2.2. Literature Review on Pre-Service Teacher Knowledge of Geometry

Previous research has reported that teachers and pre-service teacher knowledge of geometry is limited. In fact, geometry is one mathematics domain in which pre-service teachers have been found to lack teaching confidence (Jones et al., 2002). For instance, Kurt-Birel et al. (2020) found that teachers failed to master the basic properties of quadrilaterals. They were therefore unable to identify similarities and differences between types of quadrilaterals, providing incorrect definitions and inadequate inclusive classifications. Similarly, Bernabeu et al. (2021) concluded that pre-service primary school teachers struggled to define 2D figures and 3D figures in an inclusive classification. The reason was their inability to properly analyse these figures, due, in turn, to the limited range of merely prototypical examples they had been exposed to. Moreover, these pre-service teachers had greater difficulties with 3D figures than with 2D figures due to insufficient specialised knowledge. Regarding teacher training programmes, Sunzuma and Maharaj (2019) highlighted that almost half the study sample displayed geometry deficiencies, potentially affecting the teaching and learning of geometry concepts. They had not been adequately trained to teach geometry during the teacher education programmes. Such programmes should thus support pre-service teachers in overcoming these difficulties.

Given the significant implications of developing curricular noticing competence in teacher education programmes, and pre-service and teachers' specific difficulties in geometry, the main study objective was as follows: characterising a PT group's noticing of a curricular material in Spain, i.e., a 3D geometry textbook activity sequence.

2.3. Curricular Noticing Conceptualisation

We used the curricular noticing conceptualisation established by Dietiker et al. (2018) who distinguished three skills: curricular attending, curricular interpreting, and curricular responding. Curricular attending refers to the ability to perceive and recognise characteristics of curricular materials. It requires the ability to extract explicit information from a specific aspect of the material before interpreting it. Curricular interpreting refers to the act of linking mathematical and pedagogical knowledge to the curricular materials, making sense of what has been identified. For its part, curricular responding involves making informed decisions about curricular materials based on the aspects previously identified and interpreted. Providing a curricular response entails selecting which activities or materials should be presented or omitted, as well as deciding which ones should be adapted to support students' learning (Amador et al., 2017).

Therefore, curricular noticing can be broken down into a three-step process: attending to the opportunities and/or limitations of the curricular materials; interpreting how these materials can create learning opportunities; and finally, deciding how to use them during the lessons (Dietiker et al., 2018).

Research on curricular noticing has begun to describe in detail how pre-service teachers notice curricular materials (Amador et al., 2017; Cavanagh & Tran, 2023; Earnest & Amador, 2019; McDuffie et al., 2018). It has been shown that when pre-service teachers compare different versions of the same activity, they succeed at identifying differences in their designs and at perceiving new teaching opportunities in the materials (Amador et al., 2017). McDuffie et al. (2018) found that although teachers attended to similar teaching material characteristics, their interpretations and subsequent decisions varied according to their orientations and the type of curriculum. Zorrilla et al. (2024) identified four different pre-service teacher profiles according to how they interpreted limitations, and how they completed an activity sequence on the part-whole meaning of the fraction concept: (1) pre-service teachers who interpreted only; (2) pre-service teachers whose activity sequence was not coherent with their interpretation; (3) pre-service teachers whose activity sequence was coherent with at least one interpreted characteristic, but had difficulties in designing the activities; and (4), pre-service teachers whose activity sequence was coherent with the interpretation of at least one characteristic and did not show any design difficulties. This latter study centred on the fraction concept, exploring the relationships between interpreting and deciding in curricular noticing. It is necessary, however, to explore curricular noticing across a greater number of specific mathematics domains, e.g., geometry.

In our study, curricular noticing is understood as knowledge-based reasoning. Indeed, pre-service teachers need to use mathematical and pedagogical mathematical knowledge to attend to, interpret and decide about the curricular materials. Given the context of our study—which used a textbook activity sequence focusing on 3D geometry—the three professional noticing skills were specifically operationalised as: (i) identifying relevant characteristics in a textbook activity sequence; (ii) interpreting the activity sequence limitations for learning about the polyhedron, prism and pyramid concepts; and (iii), deciding how to complete the activity sequence considering the interpreted limitations.

According to prior research, adopting a theoretical lens is one way of fostering pre-service teachers' noticing skills in teacher education programmes (Fernández & Choy, 2019; Ivars et al., 2018). The use of theoretical lenses informed by Mathematics Education research

on the teaching and learning of specific mathematical concepts has indeed been shown to enhance pre-service teachers' noticing (Fernández & Choy, 2019). Such theoretical tools provide teachers with the necessary knowledge to reflect on and respond to a teaching-learning situation (Ivars et al., 2018). In our study, we provided pre-service teachers with a theoretical lens to analyse the textbook activity sequence. The theoretical support contained research-based information on the teaching and learning of 3D geometry in Primary Education (e.g., Battista, 2012; Bernabeu et al., 2021).

2.4. Our Study: Objective and Research Questions

The study objective was to examine how PTs interpreted various characteristics of a 3D geometry activity sequence—namely, attributes, modes of representation, and geometrical processes while identifying limitations in supporting students' learning of the polyhedron, prism, and pyramid concepts, and proposed modifications to the sequence to address these limitations using a theoretical lens.

The research questions were formulated as follow:

- How do PTs interpret characteristics of a textbook activity sequence while also recognising its limitations in supporting students' learning of polyhedron, prism, and pyramid concepts? (RQ1)
- How do PTs complete the activity sequence to address these limitations? (RQ2)

Our study contributes to the existing literature by unravelling how, using a theoretical lens, PTs interpret and decide about a sequence of activities on a specific mathematical concept—specifically the polyhedron, prism and pyramid. This approach allowed to characterise PTs curricular noticing in the specific domain of 3D geometry using a theoretical lens to frame the curricular materials.

3. Method

This mix-method study focused on how PTs interpreted and completed a sequence of activities (i.e., made decisions), exploring the coherence between their interpretations and decisions. We quantified the frequencies of the PTs' interpretations and decisions and then, moving on to a qualitative approach, we conducted an inductive analysis to generate categories based on inter-researcher agreement.

The participants, context, instrument, and the performed analysis are all described below.

3.1. Participants and Context

A total of 146 PTs were enrolled in the Teaching and Learning of Mathematics in Primary Education course at the University of Alicante (Spain), which is part of the third year of the Primary Education Teacher Degree. The main objective of this course is to develop professional mathematics teaching competences, such as interpreting students' mathematical thinking, analysing curricular materials, and planning lessons in different mathematical domains (e.g., geometry). The course was implemented by two authors of this work. The PTs had previously completed two other mathematics courses: one related to number sense in the first year of the degree, and the other to geometric sense in the second year. These previous courses were oriented towards mathematical content.

The PTs participated in a teaching module described in the following section. Of note, excluded from the study were the PTs who did not answer instrument questions 2 or 3 (see section below) since one research question was to explore how PTs completed the activity sequence to address the limitations. Consequently, the final dataset comprised 85 participants who provided complete answers to both questions.

3.2. Instrument: Task and Theoretical Lens

The PTs participated in an 8 h teaching module on primary school geometry teaching-learning which was part of the course. In this module, the PTs solved a task consisting of five activities drawn from a Year 5 (students aged 10 to 11 years) Spanish 3D geometry textbook (Figure 1) as well as a set of questions to guide them in their analysis of the activity sequence. The PTs solved this task individually during a two-hour lesson which was part of the teaching module.

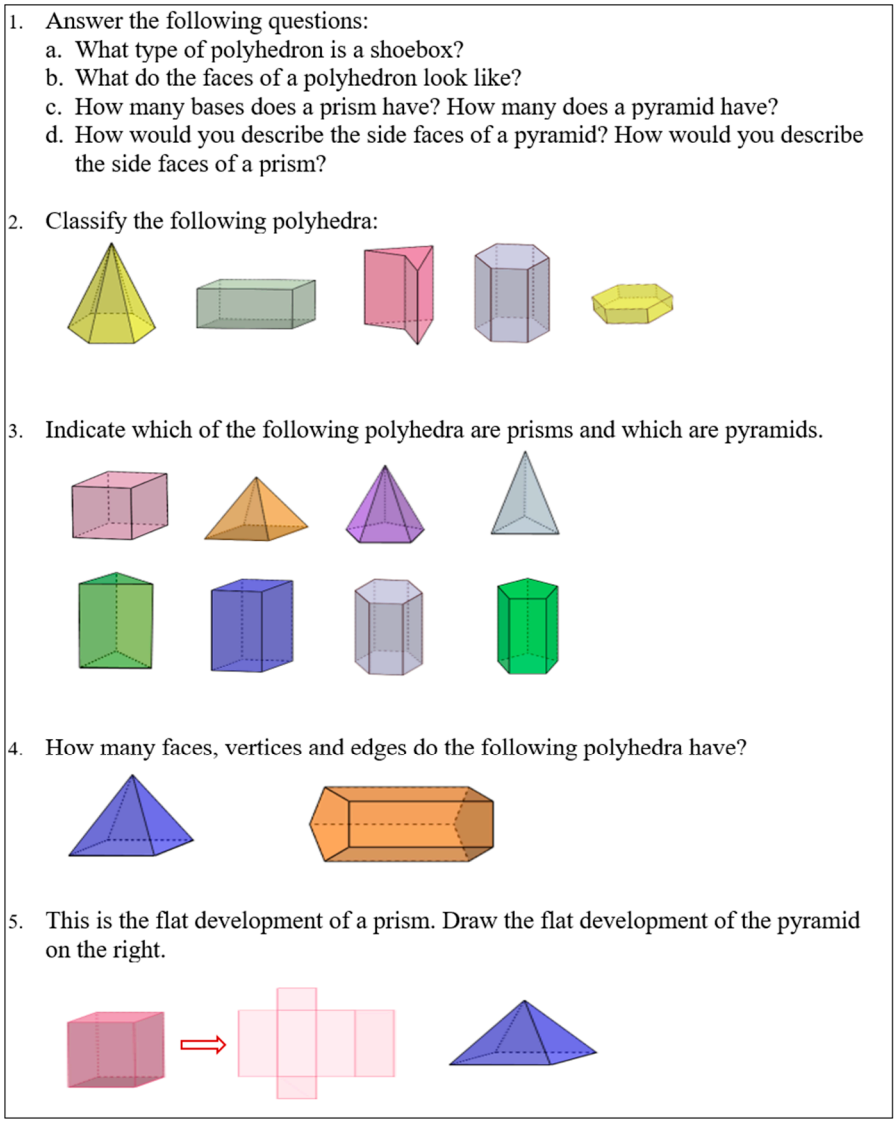

1. Answer the following questions:
   a. What type of polyhedron is a shoebox?
   b. What do the faces of a polyhedron look like?
   c. How many bases does a prism have? How many does a pyramid have?
   d. How would you describe the side faces of a pyramid? How would you describe the side faces of a prism?
2. Classify the following polyhedra:
3. Indicate which of the following polyhedra are prisms and which are pyramids.
4. How many faces, vertices and edges do the following polyhedra have?
5. This is the flat development of a prism. Draw the flat development of the pyramid on the right.

Figure 1. Textbook activities (translated into English from Ferrero de Pablo et al., 2018, pp. 178–179).

The questions provided in the task were related to curricular noticing:

- For each activity, indicate: (i) Which geometrical process/es are used (e.g., recognising, constructing, classifying, etc.); (ii) The attributes of each 3D figure; and (iii) The modes of representation.
- Considering the activity sequence: (i) What other 3D figure attributes could be added to help students acquire the polyhedron, pyramid and prism concepts? (ii) What other geometrical processes could be added? and (iii), What other modes of representation could be introduced?
- Complete the activity sequence (a minimum of 3 activities) to support the students' understanding of the concepts of polyhedron, prism, and pyramid.

The set of questions directed the PTs' attention towards three important activity characteristics: attributes, geometrical processes and modes of representation. The activity sequence involved the geometrical processes of recognising (activities 1 and 3), classifying (activity 2), analysing (activities 1, 4 and 5), and constructing (activity 5). Regarding the attributes, all polyhedra were prisms or pyramids (only a concave prism appeared and there was no oblique prism or pyramid). Therefore, most polyhedra were prototypical prisms or pyramids. Concerning the modes of representation, the activity sequence comprised the graphical and written modes of representation. PTs could identify some limitations in the activity sequence: the absence of oblique pyramids or prisms, polyhedra that were neither prisms nor pyramids, or non-examples of polyhedra. Furthermore, they could add defining or measuring processes and concrete (manipulative) materials or oral contexts as modes of representation.

The PTs were also provided with a theoretical lens, i.e., a theoretical document containing information from mathematics education research relating to the teaching and learning of 3D geometry in Primary Education (e.g., Battista, 2012; Bernabeu et al., 2021). The document included information on geometrical processes (recognising, analysing, classifying, defining, constructing, or proving), modes of representation (verbal, symbolic, concrete or graphic), and the importance of using a large variety of examples with different 3D figure attributes (e.g., convex, concave, oblique, right, etc.), as well as non-examples. The mathematical content on 3D figure attributes was provided during the second-year course on geometric sense.

3.3. Data and Analysis

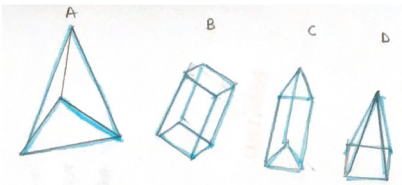
The study data consisted of the PTs' answers to questions 2 and 3. The analysis was conducted in three phases. Phase 1 focused on the PT interpretations (RQ1). In this phase, four researchers individually identified the attributes, the modes of representation, and the geometrical processes that were signalled by the PTs as missing in the textbook activity sequence and that would have helped students to understand polyhedron, prism and pyramid concepts. Seven categories emerged from this inductive analysis (Strauss & Corbin, 1990) regarding the following attributes: "concave/convex"; "oblique/right"; "regular/irregular"; "the number of base sides"; "non-prototypical"; "the use of non-examples"; and "attributes that appear in the sequence or blank answers". For their part, four geometrical process categories arose: "measuring", "proving", "defining", and "geometrical processes that appear in the sequence or blank answers". Finally, three representation mode categories emerged: "manipulatives", "oral" and "modes of representation that appear in the sequence or blank answers". The categories "attributes that appear in the sequence or blank answers", "geometrical processes that appear in the sequence or blank answers" or "modes of representation that appear in the sequence or blank answers" indicated that the PTs highlighted characteristics that appeared in the sequence or did not answer the specific question. These PTs therefore failed to interpret the activity sequence limitations regarding the specific characteristic. The different categories were refined and discussed until the four researchers reached a unanimous agreement (Table 1 displays an example of this analysis).

Phases 2 and 3 are related to RQ2. Phase 2 focused on identifying the characteristics used by PTs in the designed sequence of activities that differed from the textbook activities (RQ2). In this phase, the four researchers individually analysed the PTs' designed activity sequences. We identified the attributes, the geometrical processes involved in the activities, and the modes of representation present in the PTs' designed sequences which differed from the textbook sequence. Again, the four researchers discussed the emerging characteristics identified until reaching a unanimous agreement (see Table 2 for an example of this analysis).

Table 1. Examples of the analysis in phase 1 (interpreting).

Characteristic	PT Answer	Description	Category
Attributes	To understand the polyhedron concept, we could propose activities to work on concavity and convexity. These attributes can also be worked on using prisms and pyramids.	PT01 identifies "concavity and convexity" as a missing attribute in the sequence	"concave/convex"
Geometrical process	Among the geometrical processes, those of recognising attributes, classifying and constructing figures are addressed. We could therefore propose activities in which students need to define figures.	PT01 identifies "defining" as a missing geometrical process in the sequence	"defining"
Mode of representation	To promote understanding, we could use concrete materials such as the polydron.	PT01 identifies a "manipulative" as a mode of representation missing in the sequence	"manipulative"
Attributes	Concave polyhedra, irregular polyhedra, polyhedra in non-prototypical positions, oblique polyhedra.	PT02 identifies "concavity, irregularity, obliquity and non-prototypical positions" as missing attributes in the sequence	"concave/convex"; "oblique/right"; "regular/irregular"; "non-prototypical"
Geometrical process	I would suggest tasks involving figure definition and construction, since most activities involve identifying and classifying.	PT02 identifies "defining" as a missing geometrical process in the sequence	"defining"
Mode of representation	Graphical representation and using concrete materials would help to support understanding.	PT02 identifies "concrete materials" (manipulatives) as a mode of representation missing in the sequence	"manipulatives"

Table 2. Examples of the phase 2 analysis (deciding).

PT Answer	Identified Characteristics Which Differed from the Textbook Sequence
PT03's answer Activity 1. Define the following figure Activity 2. Construct a heptagonal right prism. Activity 3. Classify the following figures into prisms or pyramids and according to the base (quadrangular or triangular):  	Attributes: number of base sides (hexagonal, triangular, etc.) Geometrical processes: defining
PT02's answer Activity 1. Draw the figure Activity 2. Define the figures that appear in activity 1. Activity 3. Classify the figures in activity 1 according to the number of the sides of their bases.	Attributes: number of base sides Geometrical processes: defining

In phase 3, we analysed the relationship and coherence between the PTs' interpretations and decisions (Rotem & Ayalon, 2024). This analysis shed light on how PTs completed the activity sequence to address the limitations. The decisions were considered coherent with respect to the interpretations when the PTs' designed activity sequences addressed the previously interpreted limitations (i.e., the categories obtained in phase 1). Consequently, in this phase, we analysed whether the PTs used what they indicated as missing characteristics (phase 1 categories) when designing the activity sequence (characteristics identified in the designed activity sequence that differed from the textbook; phase 2). Again, the four researchers individually analysed the coherence of each characteristic (attributes, geometrical processes, and modes of representation) until reaching a unanimous agreement (Table 3). Of note, a PT's decision could be coherent regarding one characteristic or more. Also detected was the fact that the PTs had difficulties in designing the activities since they committed certain inaccuracies. For example, they proposed activities which did not involve 3D geometry, or suggested activities without specifying the figures. We also verified whether any PTs added characteristics that differed from the textbook sequence but had failed to identify in their previous interpretation.

Table 3. Example of the analysis in phase 2 (relationship between interpreting and deciding).

	Relating Interpreting and Deciding	Coherence	Other Characteristics and PTs Difficulties
PT02	PT02 identifies "defining" as a missing geometrical process in the sequence and designs an activity of defining	Coherence with: geometrical processes	This PT adds an attribute "number of base sides" that had not previously been identified. Any inaccuracy was detected in the designed activity sequence.

4. Results

The study findings are described in the two subsections below. Section 4.1 summarises how the PTs interpreted the textbook activity sequence (RQ1). Section 4.2 describes how they completed the activity sequence to address the interpreted limitations (RQ2).

4.1. PT Interpretations

Table 4 shows the number of PTs who interpreted the different characteristics of the textbook activity sequence (attributes, geometrical processes, and modes of representation). All 85 PTs interpreted at least one characteristic. A total of 20% (17 PTs) interpreted only one characteristic, 36.5% interpreted two characteristics (31 PTs), and 43.5% interpreted the three characteristics (37 PTs). The fact that 56.5% of the PTs failed to recognise limitations in the three different characteristics (geometrical processes, modes of representation and attributes) revealed interpretation difficulties.

Table 4. Distribution of PTs according to the interpreted activity sequence characteristics.

Attributes	Number of PTs
Attributes, geometrical processes and modes of representation	37
Attributes and geometrical processes	11
Geometrical processes and modes of representation	5
Attributes and modes of representation	15
Attributes	10
Geometrical processes	1
Modes of representation	6

A larger number of PTs were capable of interpreting the attributes (73 PTs, 85.9%), followed by modes of representation (63 PTs, 74.2%), and finally the geometrical processes (54 PTs, 63.5%). Therefore, the PTs were more adept at recognising activity sequence limitations concerning attributes than those associated with modes of representation or geometrical processes.

4.2. PT Decisions to Address the Limitations

Table 5 shows that the activity sequence designed by one PT group (17 PTs, 20%) was not coherent with the interpreted characteristics. It also illustrates the coherence of the activity sequence of a group of 68 PTs (80%) regarding at least one interpreted characteristic (modes of representation, geometrical processes or attributes). Therefore, most PTs designed an activity sequence which was coherent with at least one interpreted characteristic, adapting and modifying curricular material to address specific interpreted limitations.

Table 5. PT distribution according to the coherence between their decisions and interpreted characteristics.

Attributes	Number of PTs
Attributes, geometrical processes and modes of representation	1
Attributes and geometrical processes	3
Attributes and modes of representation	7
Geometrical processes and modes of representation	3
Attributes	38
Geometrical processes	7
Modes of representation	9

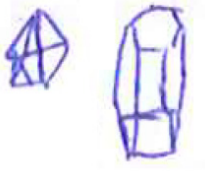
Of the 68 coherent PTs, 56.5% (48 PTs) designed an activity sequence which was coherent with the interpreted attributes, 16.5% (14 PTs) designed an activity sequence according to the geometrical processes interpreted, and 23.5% (20 PTs) designed an activity sequence which was coherent with the mode(s) of representation. Therefore, when designing an activity sequence aimed at enhancing student understanding, the PTs appeared to focus more on adding attributes and less on modes of representation or geometrical processes.

Furthermore, given that the PTs could design an activity sequence considering at least one characteristic, Table 5 also shows that only 1 PT (1.2%) was coherent regarding the three characteristics, while 13 PTs (15.3%) were coherent with two characteristics, and 54 PTs (62.3%) were coherent with one characteristic. The two subsections below present examples of the PTs' decision characteristics that were not coherent with their interpretation and the characteristics of the PTs' decisions that were coherent with their interpretations.

4.2.1. PTs Who Were Not Coherent with Their Interpretations

This group of 17 PTs designed an activity sequence but failed to consider any of the interpreted characteristics. However, we identified different characteristics relating to the PTs' answers within this group. A total of 10 PTs designed an activity sequence integrating other missing characteristics of the textbook activity sequence that were not interpreted. For instance, PT57 gave the following interpretation of the textbook activity sequence: "we could add more concave and oblique figures, activities based on edges, vertices and faces and on drawing figures, and the use of manipulative material". As can be observed, this PT interpreted the absence of manipulative material as a textbook limitation (thereby interpreting only one characteristic—the modes of representation). Yet, in the designed activity sequence (Figure 2), this PT was not coherent with this latter interpretation, failing to propose the use of manipulatives. Instead, the PT added a different missing characteristic in the designed activities: regularity (activity 2, Figure 2).

Activity 1. Draw a quadrangular pyramid and a prism.
Activity 2. Classify the following figures according to regular and non-regular.



Activity 3. Indicate the part of a polyhedron.

Figure 2. PT57's designed activity sequence.

Five of the 17 PTs also had difficulties in designing the activity sequence since they did not provide specific activities to work the polyhedron, prism and pyramid concepts. These PTs proposed activities which did not involve 3D geometry or proposed activities without specifying the figures, therefore the activities could not be solved. For example, PT27 interpreted "we can add concave, oblique and irregular figures, measuring activities and the use of manipulatives and oral answers". Therefore, this PT interpreted the limitations regarding the three characteristics (adding other attributes, geometrical processes and modes of representation). However, this PT had difficulties in the designed activity sequence, because the sequence of the three proposed activities did not allow working on 3D geometry. Figure 3 presents the PT's first designed activity as an example.

Identify the rectangles in the following figures:

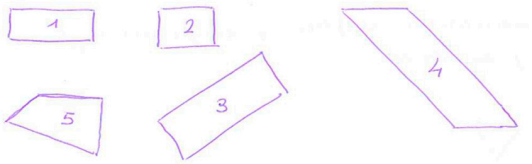


Figure 3. An example of the activities designed by PT27.

Finally, 2 PTs failed to design any activity sequence with a new characteristic. This was the case of PT62 who interpreted the attributes and geometrical processes. This PT expressed the intention to add "concavity and regularity and the geometrical processes of defining, and measuring". In the designed activity sequence (Figure 4), the PT's decision lacked coherence with the given interpretation because the interpreted attributes and processes were not integrated into the decision. Moreover, this PT did not consider any other missing textbook sequence characteristics.

Activity 1. A polyhedron has 2 hexagonal faces and the others are triangles. Indicate the number of edges of this polyhedron.
Activity 2. A polyhedron has 7 faces, 4 are pentagons and 3 quadrilaterals. How many vertices does this polyhedron have? And how many edges does it have?
Activity 3. Draw the following figure. What is the name of the figure?

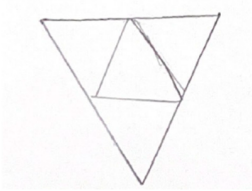


Figure 4. PT62's designed activity sequence.

4.2.2. PTs Who Were Coherent in Relation to Their Interpretation

This group of 68 PTs designed an activity sequence that was coherent with at least one of the interpreted characteristics (modes of representation, attributes or geometrical processes). We identified different PT answer characteristics within this group: 13 PTs addressed some of the interpreted characteristics accurately; 30 PTs also addressed other, previously non-interpreted missing characteristics; 25 encountered difficulties in designing the activity sequence, as they incorporated at least one activity that did not allow for engagement with the concepts of polyhedron, prism and pyramid.

For example, PT53 deemed that the textbook activities did not address the "regular/irregular", "right/oblique", or "non-prototypical" attributes, and also noted the absence of the "manipulative" mode of representation. As shown in Figure 5, PT53 incorporated these characteristics when designing the proposed activity sequence. In activity (a), the regular and irregular attributes were addressed by requiring the drawing of regular and irregular 3D figures. This attribute was further emphasised in activity (b) by asking to construct various regular, irregular, and non-prototypical polyhedra using origami. Furthermore, the use of the manipulative representation mode was added. Activity (c) explicitly targeted the regularity and right/oblique attributes which had previously been interpreted by this PT.

(a) Draw three regular polyhedrons (a pyramid, a prism, and another) and corresponding irregular ones in your notebook.

(b) Construct the following (properly measuring the paper with a ruler) (origami)
- Quadrangular prism
- Tetrahedron
- Icosahedron
- Irregular Polyhedron
- Truncated pyramid

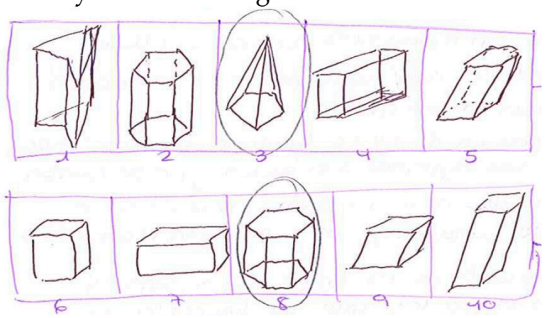
(c) Analyse the figures you have constructed and indicate the type of polyhedron. Also indicate how many faces they have, whether they are regular or irregular, right or oblique.

Figure 5. PT53's designed activity sequence.

PT21 interpreted the three characteristics highlighting that other attributes (oblique figures), other geometrical processes (measuring), and other modes of representation (oral and manipulative materials) could be added. In the activity sequence designed (Figure 6), this PT addressed the highlighted attribute (oblique/right) in activity 1, but also added another "defining" geometrical process (activity 3) that had not been interpreted previously.

The group of PTs that had difficulties in designing the activities proposed non-3D geometry tasks or proposed activities without specifying the figures. This was the case of PT41 who interpreted only the non-use of the manipulative materials as limitations of the textbook activity sequence. In the designed activity sequence (Figure 7), this PT introduced the manipulative materials of chopsticks and modelling clay (thereby displaying coherence with the interpretations made). However, one of the designed activities did not focus on 3D geometry (activity 2), thereby revealing activity design difficulties.

Activity 1. Circle the figure that is different from the rest and explain why.



Activity 2. Draw 2 pyramids with a different base and 2 prisms with a different base. Indicate the number of vertices, faces and edges.
Activity 3. Define the prism and pyramid. Then define the pyramids and prisms you have drawn in the previous exercise.

Figure 6. PT21's designed activity sequence.

Activity 1. Using chopsticks and modelling clay, construct a pyramid and a prism.
Activity 2. Use the geoboard to form three different quadrangular polygons.
Activity 3. Indicate 2 characteristics of prisms and 2 characteristics of pyramids. Draw a pyramid and a prism to observe the characteristics you have indicated.

Figure 7. PT41's designed activity sequence.

5. Discussion and Conclusions

The present study examined how PTs interpreted different characteristics of an activity sequence, recognising limitations in its ability to support students' mathematical understanding of the polyhedron, prism and pyramid concepts. It also centred on how the PTs completed the activity sequence to address the limitations, examining the coherence between the PTs' interpretations and decisions.

Regarding curricular interpreting, the 85 PTs succeeded at identifying at least one characteristic concerning the attributes, geometrical processes or representation modes. Therefore, all PTs interpreted some textbook activity sequence limitations. These results suggest that a theoretical lens and guiding questions help PTs to interpret activity limitations regarding the understanding of the polyhedron, prism, and pyramid concepts. Nevertheless, other factors can also influence the PTs' interpretations, such as their own mathematical and pedagogical knowledge.

Our results also showed that a larger number of PTs successfully interpreted the attributes, followed by the modes of representation and the geometrical processes. Interpreting geometrical processes, therefore, constitutes the most challenging characteristic for PTs to identify.

On the subject of curricular decisions, 17 out of the 85 PTs failed to be coherent with respect to any of the interpreted characteristics. These PTs struggled to connect their interpreted activity sequence limitations with the decisions they made. These results are consistent with previous findings according to which PTs are able to make accurate interpretations but have difficulties with decision-making (Barnhart & van Es, 2015; Ivars et al., 2020). In these previous studies, PTs successfully interpreted students' mathematical thinking but later showed difficulties in using their interpretations to suggest actions focused on students' mathematical thinking. The present work allowed to extend this result to curricular noticing: The PTs were able to interpret textbook activity sequence limitations, but a subgroup had difficulties in introducing the identified limitations into their proposed activity sequence. Interviews should be conducted to explore the possible reasons for this. Practical implications can nevertheless be drawn: to provide PTs with the

necessary experience and knowledge, teacher training programmes should focus more on incorporating the type of task used in this study.

In our study, 80.0% of the 85 PTs designed an activity sequence which was coherent with at least one interpreted characteristic, successfully adapting and modifying an activity sequence to enhance student understanding. This outcome highlights that the activity sequence task, supported by guiding questions and a theoretical lens, may help PTs to design activity sequences that are specifically directed towards supporting student understanding (Fernández et al., 2018; Fernández & Choy, 2019). Moreover, our results revealed that PTs focused more on adding attributes to their activity sequence than on promoting other modes of representation or geometrical processes. This finding suggests that PTs tend to prioritise certain task characteristics over others. Specifically, the PTs focused more on perceptual characteristics (attributes), emphasising the important role of visualisation. However, to better understand why PTs prioritised certain characteristics over others, it would be necessary to conduct interviews or request written justifications for their decisions. Our results on the priority given to specific characteristics support that of Zorrilla et al. (2024) and Llinares et al. (2025). In these latter studies, when the PTs interpreted and decided about an activity sequence related to fractions, they preferred to change the activity type than to modify the mathematical elements or the representation modes. Consequently, further research is needed to explore the reasons underlying these differences across mathematical domains.

The results revealed both a non-coherent group and a coherent group of PTs with their interpretations—the coherent group integrated at least one of the interpreted characteristics into their activity sequence design. Within each group, we identified certain characteristics regarding how they completed the activity sequence to address the limitations. One PT group addressed other, non-interpreted missing characteristics of the textbook activity sequence. Another group had difficulties in the design and failed to provide specific 3D figures—or they designed activities which were unrelated to the polyhedron, prism or pyramid concepts. This latter PT group had difficulties in designing 3D geometry activities. As suggested by Bernabeu et al. (2021), PTs often encounter obstacles when working with 3D figures, thereby suggesting that their understanding of 3D figures may be superficial and limited. Such an understanding is, moreover, probably based on memorised definitions or properties (Battista, 2012), rather than on relevant attributes and geometrical processes. This result is significant: indeed, the PTs had completed a geometry course (including 3D geometry contents) in the previous year, yet despite this, they continued to exhibit mathematical knowledge gaps. According to prior studies on the noticing of student mathematical thinking, PTs struggle to interpret and decide when they fail to master the necessary mathematical content knowledge (Buforn et al., 2022; Ivars et al., 2020). The present study thus extends the scope of this finding to the PTs' curricular noticing.

Our study contributes to the curricular noticing literature and has important implications for the design of mathematics education courses. On the one hand, empirical evidence was obtained on how PTs interpret and decide to complete a textbook 3D geometry activity sequence. The study results offer specific insights into PT interpretation and decision-making processes when interpreting limitations and proposing modifications to complete a textbook activity sequence. Textbooks are the most widely used educational resource to prepare mathematics lessons (Blanco et al., 2021; Glasnovic Gracin, 2018). Moreover, the textbooks often present a range of distinct shortcomings (Barrantes-López et al., 2014; Yang et al., 2017). Teacher educators must therefore engage PTs in critically identifying textbook strengths and weaknesses, as well as in designing activities that address compensate those limitations. The findings of the present study help teacher educators to design mathematics courses that empower PTs to use their knowledge, thereby making more effective

choices to promote student learning, particularly by considering the limitations detected in textbook activities.

The characteristics that were identified of the pre-service teachers' coherence between interpretation and decision-making offer valuable insights into which knowledge domains require deeper attention in teacher education programmes (Sunzuma & Maharaj, 2019). More specifically, initial mathematics teacher education programmes should integrate additional curricular noticing tasks, such as the task and theoretical tool used in our study, as this would allow PTs to develop curricular noticing earlier in their teaching careers.

To finish, the results cannot be generalised to other educational contexts since the PTs' noticing can be influenced by a country's educational context, or even by the specific features of the training programme. As mentioned above, the present research could be strengthened in future studies by including interviews that explore the PTs' justifications. Moreover, future works could focus on which factors facilitate or hinder interpretation and decision coherence.

Author Contributions: Conceptualization, C.F., Z.O.-L., A.S. and M.B.; Methodology, C.F., Z.O.-L., A.S. and M.B.; Formal analysis, C.F., Z.O.-L., A.S. and M.B.; Writing—original draft, C.F., Z.O.-L., A.S. and M.B.; Writing—review & editing, C.F., Z.O.-L., A.S. and M.B.; Project administration, C.F.; Funding acquisition, C.F. All authors have read and agreed to the published version of the manuscript.

Funding: This research was supported by the project CIAICO/2021/279 funded by the Generalitat Valenciana (Conselleria d'Educació, Cultura i Sport, Spain).

Institutional Review Board Statement: The study was conducted in accordance with the Declaration of Helsinki, and approved by the Ethical Committee of University of Alicante (protocol code: UA-2023-02-06_2; date of approval: 1 March 2023).

Informed Consent Statement: Informed consent was obtained from all the participants.

Data Availability Statement: To maintain the privacy of the participants, the data are not publicly available.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

3D three-dimensional
PTs pre-service primary school teachers

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